



Non-neoplastic Intestinal Disease

Pathology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 9 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.

EASY

Which inflammatory bowel disease characteristically shows skip lesions?

- A. Ulcerative colitis
- B. Crohn disease
- C. Both equally
- D. Neither — skip lesions are not a feature of IBD
- E. Ischemic colitis

2.

EASY

Which IBD shows non-necrotizing granulomas on biopsy?

- A. Ulcerative colitis
- B. Crohn disease
- C. Both equally
- D. Neither
- E. Only in the presence of fistulae

3.

EASY

Hirschsprung disease occurs in approximately how many live births?

- A. 1 in 500
- B. 1 in 5,000
- C. 1 in 50,000
- D. 1 in 500,000
- E. 1 in 5,000,000

4.

EASY

In Hirschsprung disease, which two nerve plexuses are absent from the aganglionic segment?

- A. Auerbach and vagal
- B. Meissner and Auerbach
- C. Meissner and sympathetic chain
- D. Vagal and sympathetic chain
- E. Auerbach and parasympathetic ganglia only

5.

MEDIUM

CASE

A colectomy specimen shows continuous inflammation beginning in the rectum with an abrupt transition to normal mucosa proximally, and numerous pseudopolyps.

Which biopsy finding would you expect?

- A. Non-necrotizing granulomas
- B. Crypt abscesses with mucin depletion
- C. Transmural lymphoid aggregates
- D. Fissuring ulceration
- E. Cobblestone mucosa

6.

MEDIUM

CASE

A 26-year-old has longitudinal ulcers with intervening segments of entirely normal mucosa, and marked transmural fibrosis on resection.

Given the depth of inflammation described, which complication follows most directly?

- A. Pseudopolyp formation
- B. Toxic megacolon
- C. Stricture and fistula formation
- D. Continuous rectal bleeding from the outset
- E. Crypt abscesses as the dominant histological feature

7.

MEDIUM

CASE

A neonate fails to pass meconium within 48 hours of birth and develops abdominal distension and bilious vomiting. Contrast enema shows a narrow-calibre distal segment with proximal bowel dilatation.

Where should the diagnostic rectal biopsy be taken from?

- A. The dilated proximal segment
- B. The narrow distal segment
- C. The transition zone exclusively, since biopsying either side alone is uninformative
- D. Biopsy is not required if imaging is diagnostic
- E. Anywhere in the colon, since ganglion cells are evenly distributed

8.

MEDIUM

Which single histological finding, if present, most strongly favours Crohn disease over ulcerative colitis?

- A. Crypt distortion
- B. Mucin depletion
- C. Non-necrotizing granuloma
- D. Mucosal ulceration
- E. Regenerative epithelial atypia

9.

MEDIUM

In Hirschsprung disease, the functional obstruction arises because:

- A. Fibrous tissue mechanically narrows the aganglionic segment
- B. Peristalsis is absent in the aganglionic segment due to failure of neural crest cell migration
- C. The muscularis propria becomes hypertrophied and rigid
- D. Meconium becomes inspissated within the ileum
- E. An autoimmune process destroys ganglion cells after birth

10.

MEDIUM

CASE

A biopsy from a patient with long-standing colitis shows disease confined to the mucosa and submucosa, without fissures or transmural fibrosis, and no granulomas are identified on multiple levels.

This depth and granuloma pattern is most consistent with:

- A. Crohn disease
- B. Ulcerative colitis
- C. Either, since depth does not discriminate them
- D. Ischemic colitis specifically
- E. Diverticular disease

11.

MEDIUM

Cobblestone mucosa, resulting from longitudinal ulcers alternating with intervening normal mucosa, is a gross feature most associated with:

- A. Ulcerative colitis
- B. Crohn disease
- C. Hirschsprung disease
- D. Ischemic colitis
- E. Diverticulitis

12.

MEDIUM

CASE

A biopsy from an area of active colitis shows crypt abscesses, crypt architectural distortion, and mucin depletion, with regenerative epithelial atypia. No granulomas are seen despite deeper sectioning.

This biopsy hallmark is most consistent with:

- A. Crohn disease
- B. Ulcerative colitis
- C. Either equally
- D. Diverticular disease
- E. Ischemic colitis

13.

MEDIUM

Which of the following is proposed to contribute to IBD pathogenesis?

- A. A single identified causative organism in all cases
- B. Altered host reaction to intestinal microbiota combined with epithelial dysfunction and aberrant mucosal immune responses
- C. Exclusively a genetic cause with no environmental contribution
- D. A known autoantibody against enterocytes, analogous to autoimmune gastritis
- E. Neural crest cell migration failure, as in Hirschsprung disease

14.

MEDIUM

Which statement about the aganglionic segment in Hirschsprung disease is correct?

- A. It is the dilated segment, since dilation reflects the underlying disease
- B. It is the narrow segment, while the dilated proximal bowel is normal but obstructed
- C. It alternates between narrow and dilated in a skip pattern
- D. It cannot be identified radiologically, only histologically
- E. It always extends the full length of the colon

15.

HARD

CASE

A resection specimen shows both transmural inflammation with fissuring ulcers in the terminal ileum and a separate, disease-free segment of jejunum in between, alongside marked subserosal fibrosis.

Which feature in this description would be unusual to find in ulcerative colitis, and why?

- A. Ulceration — UC never ulcerates
- B. Involvement of the terminal ileum — UC is a colonic and rectal disease, not one that produces this distribution with skip areas
- C. Fibrosis — UC never causes fibrosis of any kind
- D. Inflammation itself — UC is entirely non-inflammatory
- E. Nothing described here is inconsistent with UC

16.

HARD

A patient has biopsy-confirmed non-necrotizing granulomas in one segment, but a separate biopsy from an inflamed but different bowel segment in the same patient shows only crypt abscesses without granulomas. Which explanation best fits the Crohn vs UC framework?

- A. This patient must have both Crohn and UC simultaneously
- B. Granulomas in Crohn disease are patchy and may not be present in every affected segment sampled, so their absence in one biopsy does not exclude Crohn if found elsewhere
- C. The second biopsy proves this is UC and the first biopsy was a laboratory error
- D. Crypt abscesses can only occur in Crohn disease, so this must be Crohn throughout
- E. Non-necrotizing granulomas, once found anywhere, confirm UC rather than Crohn

17.

HARD CASE

A neonate has delayed passage of meconium. Rectal biopsy from the visibly dilated segment shows normal ganglion cells with normal nerve fibre morphology.

What is the most appropriate interpretation of this result?

- A. This excludes Hirschsprung disease entirely, since ganglion cells anywhere in the colon are normal
- B. This is consistent with Hirschsprung disease — the dilated segment is expected to show normal ganglion cells, and the diagnostic biopsy should be repeated from the narrow distal segment
- C. This confirms Hirschsprung disease, since the dilated segment should be aganglionic
- D. This indicates a different diagnosis entirely, since Hirschsprung always causes global aganglionosis
- E. No further biopsy is needed regardless of the clinical picture

18.

HARD

Which statement correctly distinguishes the fibrosis seen in Crohn disease from findings in ulcerative colitis?

- A. Both diseases cause equivalent transmural fibrosis
- B. Crohn causes marked transmural and subserosal fibrosis, reflecting its transmural inflammation; UC, being mucosal/submucosal, does not characteristically show this
- C. UC causes more fibrosis because it is continuous rather than patchy
- D. Fibrosis is a feature of neither disease and instead indicates ischemic colitis
- E. Fibrosis in Crohn is limited strictly to the mucosal layer

19.

HARD CASE

Two neonates present with delayed meconium passage and abdominal distension. In one, contrast enema shows a normal-calibre colon throughout with delayed transit. In the other, there is a clear narrow-calibre distal segment with proximal dilatation.

Which statement is most accurate regarding the second infant compared with the first?

- A. Both infants have Hirschsprung disease, only the imaging pattern differs
- B. The second infant's pattern is classic for Hirschsprung disease; the first infant's pattern is less typical and warrants a broader differential before assuming aganglionosis
- C. The first infant's pattern is diagnostic of Hirschsprung disease and the second is not
- D. Neither pattern is consistent with Hirschsprung disease
- E. Rectal biopsy is unnecessary in either infant if contrast enema has been performed

20.

HARD CASE

An adult patient with a 15-year history of colitis, previously diagnosed and treated as ulcerative colitis based on continuous rectal-onset disease, undergoes repeat colonoscopy after developing a perianal fistula. New biopsies from a previously unsampled area of terminal ileum show transmural lymphoid aggregates and a non-necrotizing granuloma.

What is the most appropriate interpretation?

- A. The original UC diagnosis stands; fistulae and granulomas can both occur in longstanding UC
- B. This new evidence should prompt reconsideration of the diagnosis toward Crohn disease, since fistulae, ileal involvement, and granulomas are Crohn features not expected in UC
- C. The new biopsy is a contamination artefact and should be disregarded
- D. Both diagnoses are now confirmed simultaneously with no need to revise management
- E. Perianal fistulae are equally common in both conditions and carry no diagnostic weight

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	Crohn disease is characterised by skip lesions; UC is continuous from the rectum.
2	B	Non-necrotizing granulomas are a Crohn feature and are absent in UC.
3	B	Hirschsprung disease occurs in approximately 1 in 5,000 live births.
4	B	Both the Meissner submucosal plexus and Auerbach myenteric plexus are absent from the aganglionic segment.
5	B	Crypt abscesses with mucin depletion are the biopsy hallmark of ulcerative colitis; the other options are Crohn features.
6	C	Crohn is transmural, so inflammation and fibrosis extend through the wall, producing strictures and fistulae — a consequence UC's superficial disease does not share.
7	B	The narrow segment is the aganglionic, diseased bowel; the dilated segment is normal bowel obstructed behind it, and biopsying the dilated segment would show normal ganglion cells.
8	C	Non-necrotizing granulomas are present in Crohn and absent in UC; the other findings occur in both.
9	B	Aganglionosis from arrested/failed neural crest cell migration removes both plexuses, so peristalsis fails — a functional, not mechanical, obstruction.
10	B	Disease confined to mucosa/submucosa without transmural fibrosis or granulomas is ulcerative colitis, not Crohn.
11	B	Cobblestoning from longitudinal ulcers plus intervening normal mucosa is a Crohn gross feature.
12	B	Crypt abscesses, crypt distortion, mucin depletion, and regenerative atypia without granulomas is the UC biopsy hallmark.
13	B	IBD pathogenesis is believed to combine altered host reaction to microbiota, epithelial dysfunction, and aberrant mucosal immune responses — no single cause is identified.
14	B	The aganglionic segment is the narrow one; the dilated proximal bowel is structurally normal but functionally obstructed.
15	B	Transmural fissuring inflammation with skip areas and subserosal fibrosis, including ileal involvement, is characteristic of Crohn disease — UC does not typically produce this distribution or involve the ileum this way.
16	B	Crohn granulomas are patchy and may be missed on any single biopsy; their absence in one sample does not exclude Crohn if present elsewhere, so multiple biopsies matter.
17	B	The dilated segment in Hirschsprung disease is expected to show normal ganglion cells — it is normal bowel obstructed behind the aganglionic segment; the diagnostic biopsy must come from the narrow distal segment.
18	B	Crohn's transmural inflammation produces marked transmural/subserosal fibrosis; UC's mucosal/submucosal disease does not characteristically show this.
19	B	A clear narrow-to-dilated transition on contrast enema is the classic Hirschsprung pattern; a uniformly normal-calibre colon with delayed transit is atypical and should prompt a broader differential (e.g. functional constipation, other motility disorders) before assuming aganglionosis.
20	B	Fistula formation, terminal ileal involvement, transmural lymphoid aggregates, and non-necrotizing granulomas are all Crohn features that are not expected in UC — this combination should prompt diagnostic reconsideration.

Pathology GIT Block · MEDucation by Mattin Majid · MCQs written to test LG3 content — not sourced from a past paper.

Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.